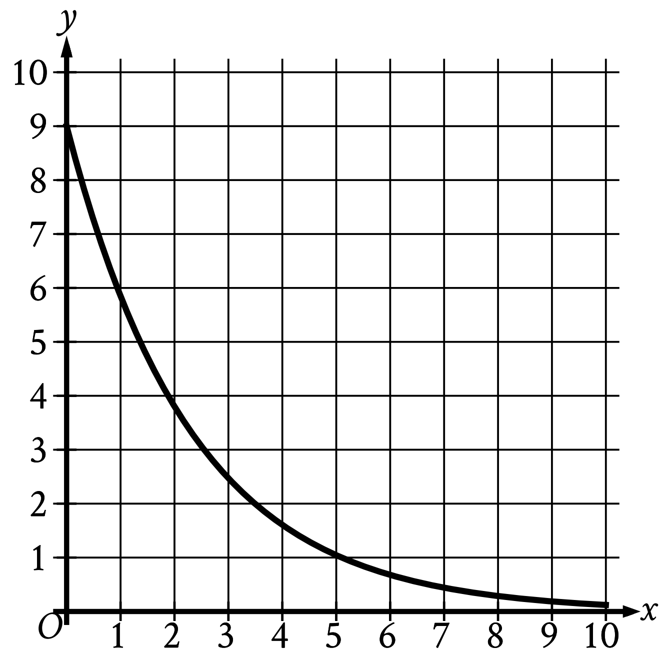
Question ID: db888cd6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question



The graph gives the estimated number of catalogs y, in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of 1992, where $0 \leq x \leq 10$. Which statement is the best interpretation of the *y*-intercept in this context?

Answer

A. The estimated total number of catalogs the company sent to its customers during the first 10 years was $9{,}000$.

B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.

D. The estimated number of catalogs the company sent to its customers at the end of 1992 was $9{,}000$.

Correct Answer: D

Rationale

Choice D is correct. The *y*-intercept of the graph is the point at which the graph crosses the *y*-axis, or the point for which the value of x is 0. Therefore, the *y*-intercept of the given graph is the point $(0, 9)$. It's given that x represents the number of years since the end of 1992. Therefore, $x = 0$ represents 0 years since the end of 1992, which is the same as the end of 1992. It's also given that y represents the estimated number of catalogs, in thousands, that the company sent to its customers at the end of the year. Therefore, $y = 9$ represents $9{,}000$ catalogs. It follows that the *y*-intercept $(0, 9)$ means that the estimated number of catalogs the company sent to its customers at the end of 1992 was $9{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.